

V2High Level Requirement Document for HOA

Project: HOA management

Time period: <3 months>

Section 1: Data Security and Privacy:

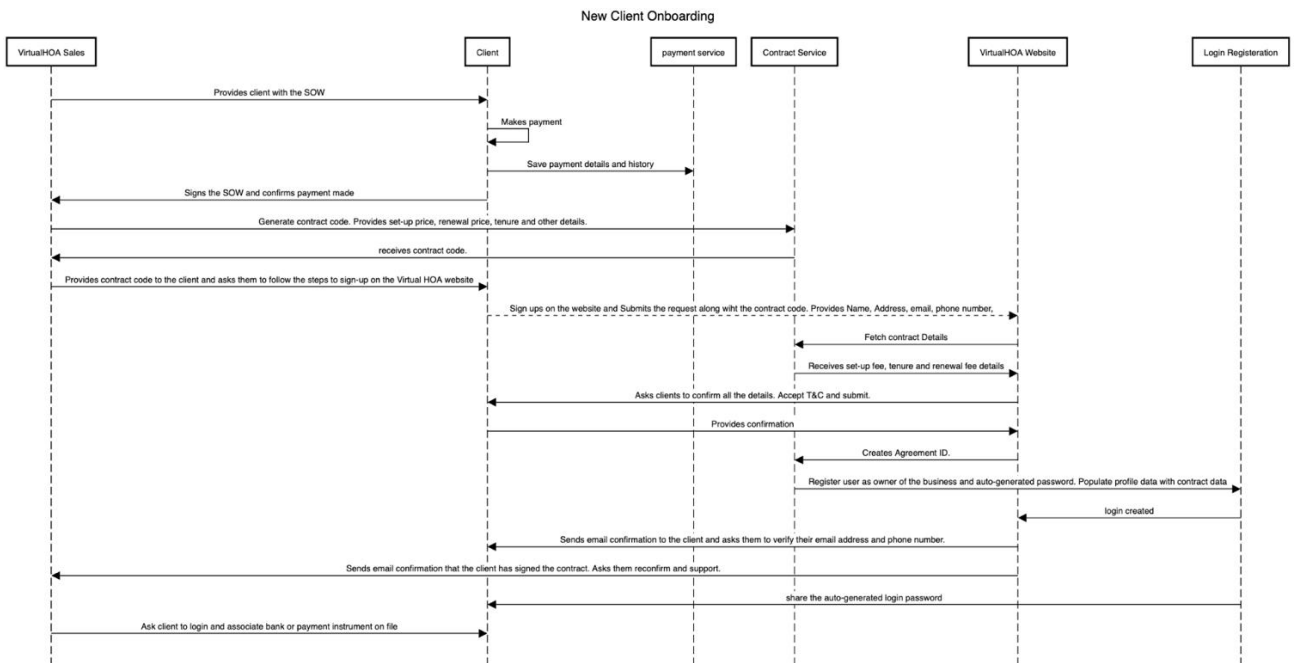
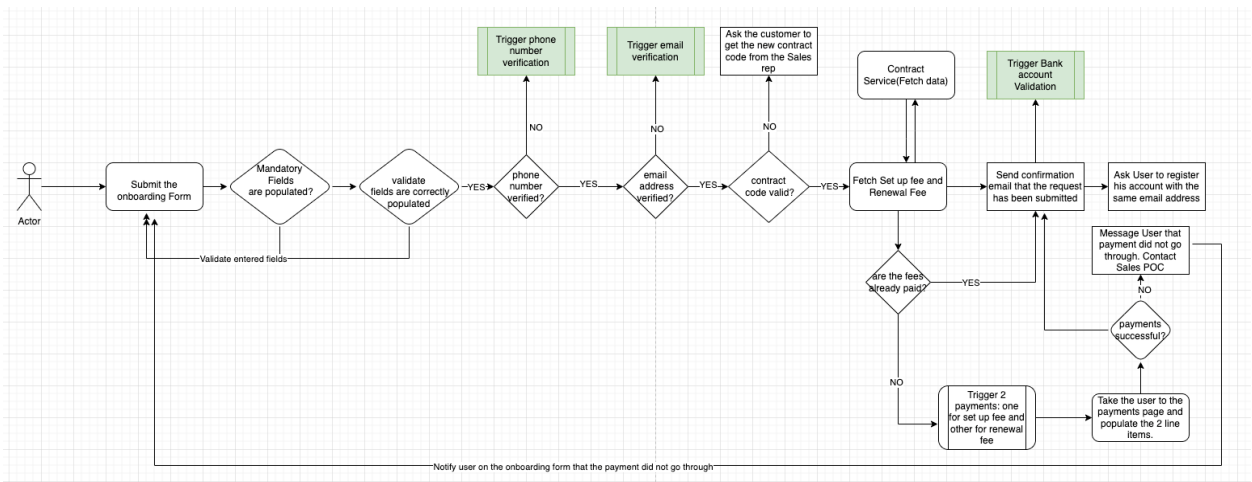
1. Email address must be verified.
2. Phone number must be verified if HOA admin/Board want to add Bank account or credit card on file.
3. Bank account and credit card information should be encrypted with only showing the last 4 digits.
4. Any time a payment is made, the transaction must be approved after SMS/OTP verified.
5. Bank accounts must be verified.
6. HOA Members and HOA Board members cannot see data of other HOA members and HOA Boards
7. HOA Admin cannot see data of other HOAs that are not linked to them
8. VHOAS team members must be authorized to review details of specific or all HOAs members/users.
9. PayPal will be used as a payment method in the US marketplace. As the product will be eventually launched WW, we must be able to customize different gateways for different marketplaces.
10. PCI 1 compliant is a must.
11. Audit history must be saved for all actions on the application. HOA admins can only view the audit history for their own HOAs. HOA Super Admin can see the audit history across all HOAs
12. All text fields should have check for 'special word check' such as slangs. (Create a service that should be called for special word check)

Section 2: HOAs sign up:

1. Vhoas sales person will contact the board members and align with them on the cost, contract and other details.
2. Sales person will then create a contract entry and provide the contract code to the Board members and ask them to submit the onboarding form. Sales person has to login to the application to create the contract.
3. Board member will sign up by filling the onboarding form (entering the contract code there). The link to the form will be provided by sales person (should be something like vhoas.com/Businessonboarding)
4. The form will require the HOA board to provide key information such as email address, contact details, phone numbers etc. (form added)
5. When they submit the onboarding form, they will be provide their bank details. We will use that bank detail to trigger the 1st month fee and the application fee.
6. Post successful payment, they will receive email confirmation that their HOA onboarding is almost complete and that they should login (using the automated password sent after

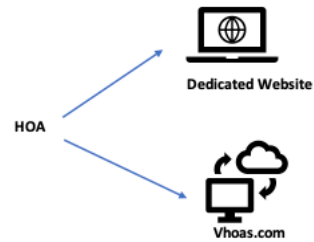
payment is successful) and update their profile and the HOA information such total number of houses in the HOA, etc.

7. Board member can then invite other board members and HOA admin. They will be required to send an email notification with a link that is specific to the HOA community.
 - a. Other board members can click on the link and follow the process
8. The board member, whose account was setup, can now invite other board members/HOA Admin to join (email invite) or provide them to join along with the community code in the registration page. Board members can find the community code in the boards tab.



Section 3: Types of HOA clients:

1. Based on website experience perspective.
 - a. Clients who use central HOA website
 - b. Clients who have their own dedicated website.
2. Based on user profile perspective:
 - a. HOA member
 - b. HOA Admin
 - c. HOA Board
 - d. HOA super Admin (this is VHOAS authorized team members)
3. Based on verified:
 - a. Non verified members: cannot do anything without verification
 - b. Verified members: can do everything based on their profile



Section 4: Central HOA vs Dedicated Website Experience

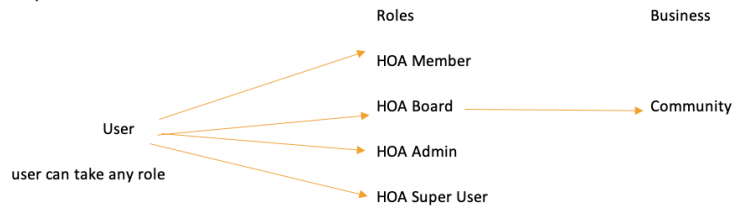
4. Two types of clients.
 - a. Clients who use central HOA website, vhoas.com
 - b. Clients who have their dedicated website.
5. Clients who use central HOA website will visit the VHOAS.COM website landing page. Landing page will provide the details about the services offered, pricing for different plans, testimonial, contact form and footer (Figma design attached). HOA members or HOA Admins can sign-in or register with CAPTCHA. Post sign-in they will be routed to their respective HOA dashboard.
6. Clients who use their own website will also have their own landing page. On their landing page, they can add their <a> own banner image add introduction <c> provide contact information and <d> ability for their customers to register and login. Register should be the public page with CAPTCHA and registration form. Sign-in will be a common sign-in form (that can be designed in #2).
7. HOA Members or HOA Admins can sign-up/login into the website. Post sign-in they will be required to a) verify their email address and b) add and verify their mobile phone number. Phone number must be verified if they want to add the bank account or download any documents from the website.
8. Post login, they will land on the dashboard page.

Section 5: Users and Roles:

1. Users: Any person trying to login into the application is considered as a user. A user can play one or many roles out of the following:
 - a. HOA member
 - b. HOA Board
 - c. HOA Admin
 - d. Super User
2. A role can be created at any time by the super admins.
3. HOA members: can make payment, submit new service request, submit violation compliance, Update profile information.

4. HOA Board: There are different types/number of Board members: President, Director, Finance, Ops Manage/submit violations, board news, new meetings, new surveys, generate reports, email communication
5. HOA Admin: Manage/Submit violations, board news, new meetings, new surveys, generate reports, email communication
6. HOA super user: Can define different types of HOA Board Members. Can configure the rights and roles of HOA Board Members and HOA Administrators. Can view/manage multiple different HOA committees.
7. Users can set time zone in which they operate. They should be able to preselect the time zone from the drop-down.

A. User Relationships



Important callouts:

- 1 Do not assume that user is same as HOA Member. Users can take any of the above role or roles
- 2 HOA board is the owner of the business i.e., the community. Hence, there will be multiple owners of the business and these owners can change over time.
- 3 HOA Admin can do everything but cannot edit the bank details or add or remove board members.
- 4 HOA super admin can do all CRUD operations across all the different clients that use our product.
- 5 In some communities, board is comprised of 5 or more people also, whereas in some communities they are comprised of just 2 people.
- 6 HOA Admin may be managing multiple different communities

Section 6: Contract Set up

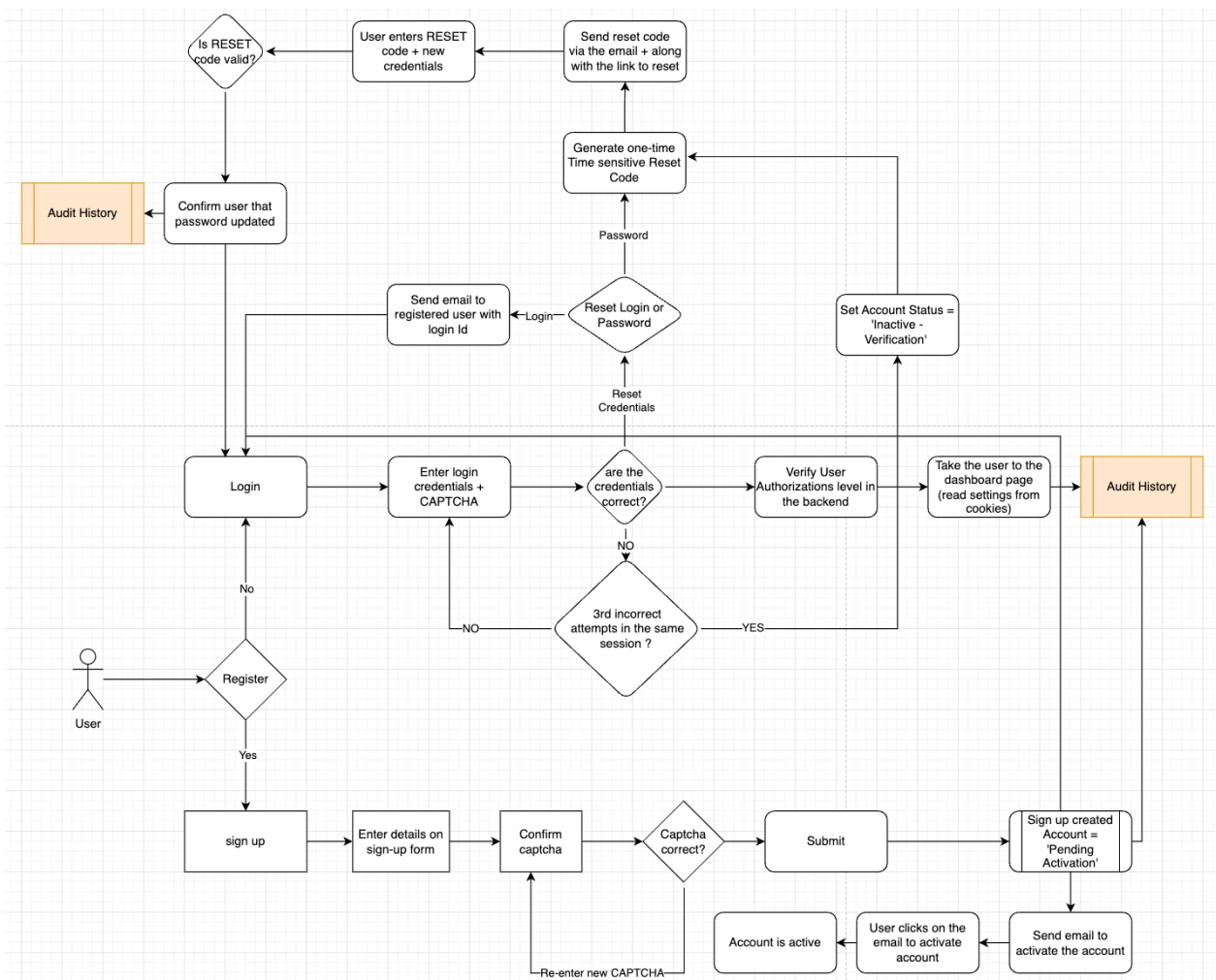
1. Sales admin will set up the contract. They will do so by logging into the HOA application.
2. Contract will include the set-up amount, renewal fee and renewal frequency.
3. Sales admin will pass the contract code to the client and ask them to register their HOA using the hidden URL and contract code.

Contract Form filled by sales team			
Has 5 different sections			
Section 1			
Sales Agent Id		autopopulated based on login	
Sales Agent Name		autopopulated based on login	
Section 2			
Date Created			
created by			
Last updated			
Last updated by			
Contract Code (auto-generated)			
Section 3			
Client First Name			
Client Middle Name			
Client Last Name			
Client Address			
City			
Zip code			
Country			
Client Phone Number			
Client Email address			
Business Name			
Business address			
Business Phone number			
Client Preferred communication Channel			
Section 4			
Plan selected			
Annual renewal Fee			
One time Set Up			
Size of the community			
Renewal cycle (monthly, Annual)			
Section 5			
Submit Button			
Save as Draft			
cancel			
Onboarding Form for clients			
1 First Name (Mandatory)			
2 Middle Name			
3 Last Name (Mandatory)			
4 Email Address (Mandatory)			
5 Phone Number (Mandatory)			
6 Your Role (Board Member, Admin)			
7 HOA Name (Mandatory)			
8 HOA address			
9 HOA contact number			
10 Contract Code (Mandatory)			
11 Add Bank Details or CC on File			
12 Captcha			

Section 7: Login/Register Experience

1. A user can login into the HOA application portal via the central HOA website (Vhoas.com) or custom HOA website.
2. Users must register with their email address and update their phone number.
3. Email notification must be sent when user completes registration.
4. Phone numbers and email addresses must be verified.
5. User should be able to reset or change their passport.

6. Users should have the capability to use 'forget password' tool to re-login in the future.
7. Audit history of login must be maintained.
8. Captcha must be integrated with registration and login in.
9. Post HOA Board sign-up, HOA Members must search for their board and submit the request to attach.
10. Post registration, a user must be assigned one of the roles at minimum. They will be assigned HOA Default role. HOA Default role will provide them no data access except asking them to choose their HOA.
11. Users can submit the request to be associated with an HOA. They should submit the request with the 'hoa pass code' and valid Identity and address proof.
12. HOA member request must be approved by HOA Admin



Section 6: HOA selection

1. HOA admins can choose one of the many HOAs that they manage.

2. When they select a particular HOA, data and reports will be shown only for that particular HOAs.

Section 7: Member profile

1. Members should be able to add or edit the details to their profiles.
2. Member profiles can be deactivated by the HOA Admin or HOA super user
3. Only HOA admin or HOA super user can re-activate the member.
4. HOA member can have more than one home in the same HOA. In this case, they should add the 2nd address (P2)
5. HOA admin will have view-only access to this page from the community detail page.
6. Members must be periodically asked to update their data as needed.
7. Email address cannot be updated/changed/deleted.

Section 8: Service Request

1. HOA members can view or edit the service request.
2. HOA Admin and HOA Super users will be able to create service request.
3. Different types of service request session:
 - a. Open
 - b. Closed
 - c. Cancelled
 - d. In-Progress
 - e. Vendor Assigned
 - f. On-Hold
4. Status of a service request can be changed by following roles:
 - a. Open -> cancelled = HOA Member
 - b. In Progress -> Vendor Assigned = HOA Board/Admin
 - c. Open -> Approved = HOA Board/Admin
 - d. *(any state) -> In-Progress = HOA Board/Admin
 - e. *(any state) -> Closed = HOA Board/Admin
 - f. *(any state) -> on-Hold = HOA Board/Admin
5. Log the time (eastern time-zone) and user details, along with the changes (Audit trail)
6. Users can see all their service request. While seeing, convert all the time to user friendly time or user time-zone selected.
7. Users can drill down and see the details of specific service request.
8. Admin or super user can assign notes, when updating the SR.
9. Service Request should have an option to be associated with a) Payment Id and b) Vendor Id.

check:

1. Can we link multiple bank accounts to the same payment gateway?
2. Payment gateway name, registration detail so that you can integrate = PayPal, visa checkout. (US)
- 3.

Section 9: Payments

There are following type of payments that must be supported.

1st Usecase: US:

- A. HOA fee (Monthly), Amentity Fee, Voilation == Community Bank Account
- B. Auto set up.

2nd Use-case: IN

- A. HOA Fee, Amentity Fee, Voilation = VHOAS Bank Account in INDIA
- B. VHOAS Bank Account transfer it to the community
- C. Auto-Set

3rd Use-case: Vendor (IN, US):

- A. HOA Admin/HOA Board : HOA will pay the vendor (vendor should have the payment page where the payment can be made).
- B. HOA Member pays to the HOA Community (Bank) -> Vendor.
 - a. HOA Member says Water leaking
 - b. HOA Admin will referral to the vendor
 - c. Vendor will give a quote (Vendor Receipt Number, Amount, Date, Service Location)
 - d. HOA member will choose a particular quote. HOA member has to pay, if they are ok with to member.
 - e.
 - f. HOA member pays
 - g. Vendor gives the invoice to the HOA Admin
 - h. HOA Admin then pays to the vendor after receiving the invoice

- 1. HOA Member paying for HOA fees: HOA Members directly pay their monthly fees to the HOA escrow account. At the time of onboarding, each HOA will provide the HOA Bank details. When a member is making a payment, they will confirm the HOA amount, HOA Name, Date and the Bank from which they are paying. Once this is done, the payment should be made.
- 2. HOA Members paying for the Amenity (optional feature)/service request: HOAs may charge their members to pay for using the HOA amenity. HOA Admin/Board can set the feature if they want to charge for the Amenity in the settings page. In this case, the money should directly go to the HOA Escow account. When a member is making a payment, they will confirm the HOA amount, HOA Name, Date and the Bank from which they are paying. Once this is done, the payment should be made.
- 3. HOA Members paying for the Violation (optional feature): HOAs may charge their members to pay for using the HOA amenity. HOA Admin/Board can set the feature if they want to charge for the violation in the settings page. In this case, the money should directly go to the HOA Escow account. When a member is making a payment, they will confirm the HOA amount, HOA Name, Date and the Bank from which they are paying. Once this is done, the payment should be made.

4. HOA Board paying to VirtualHOA (our company): HOA Board will also make set-up fee and monthly/annual payments to VirtualHOA (vhoas). This money will directly go to VirtualHOA bank account.
5. HOA Board to the Vendors: HOA Board will pay to vendors based on the work/invoice that the vendor provides. This payment will be done on the vendor portal and we do not have any work regarding this.
6. HOA Board to the Utilities: HOA Board will pay the usage fee to utility company for gas, electricity, water etc. This payment will be done on the vendor portal and we do not have any work regarding this.
7. HOA members should be able to set up recurring payments.
8. Data should be tokenized and stored with external gateway providers such as paypal, visa checkout etc.
9. Members should be able to see the history and details of all payments.
10. Confirmation email should be sent to the HOA member and HOA Board, when the payment is successful, when payment is due (2 weeks, 1 week, 1 day in advance).
11. Payments TABLE must maintain the reason for which the payment has been made.

Section 10: Amenity Booking

1. Members should be able to book the amenity that they want, if the amenity is available.
2. Allow members to choose the date and time when the amenity needs to be booked. Each day there will be two time-slots provided to book the amenity. Timeslot 1: 8am to 2pm and timeslot 2: 2pm to 8pm (everyday)
3. There should be an option for HOA members to pay a fee, when they book an amenity. This option can be configured by the HOA Board/Admin from the community settings page.
4. HOA members and HOA Board should be able to cancel the booking. Payment should be returned if the booking is cancelled. (in future, HOA Board should be able to decide if the payment is non-refundable).
5. An email notification should be sent out to the HOA members and HOA board, when the amenity is booked with the status a) payment due by date xx or b) Booking confirmed.
6. Calendar slot should not be available for the other member to book.
7. Handle Race condition i.e., when two users are trying to book the same amenity at the same time:
 - a. Before confirming, check if the calendar slot is again available. If available, block that slot (so that other parallel threads can't book it) and process the payment.
8. An email notification should be sent out when the payment is made.

HOA Member -> submits the amenity booking request -> HOA Admin approves → HOA members makes the payments -> HOA member receives email notification of the booking confirmation

Section 11: Violations

1. HOA Board members can create violation. Violation type, Violation Description, Violation Fee, Violation date, Violation Due date (X days)

2. HOA Admin/board can submit violations for a HOA member/ house (members can have multiple houses).
3. An email notification and an SMS (if set up), should go to the HOA member registered phone number and email address
4. Violation form must contain the details of the violation including date the violation, details of violation, attachment, Fine if any.
5. When the violation form is submitted, a record in the application should be created (figma screen created)
6. Violations can have following status: Open, Closed, Paid, Cancelled.
7. HOA members only have read access view of the violation.
8. Members will have 30 days to dispute the violation. They can do so by selecting the violation and hitting the dispute button/ Appeal button.
 - a. On the dispute screen, they can add a description and the attachments.
9. HOA Board will have 30 days to address the dispute.
10. Once the dispute has been addressed by the HOA board, it can no longer to re-disputed.
 - a. Members who want to further re-dispute, should be asked to contact the HOA board directly.
11. HOA members should be able to pay the fee for the violation. This option can be configured by the HOA Board/Admin from the community settings page.

Section 12: Association Details

1. Each HOA Board/Admin should be able to provide details of the HOA Board.
2. HOA board/Admin should be able to upload documents
3. HOA Board/Admin should be able to share the documents with the HOA members
4. HOA Board/Admin/Member should be able to download the document.
5. HOA member/Board/Admin should be able to see when the document was uploaded.
6. All documents must be stored in the cloud.

Section 13: FAQs

1. HOA Board/Admin should be able to create new questions and answers.
2. HOA board/Admin should be able to add links to the existing documents for members to review.
3. HOA members can only read the FAQs.
4. Should support pagination i.e., each page should show not more than 10 FAQs. Members can click to go to the next page.

Section 14: Setting Page

1. HOA Board/Admin can add and update which tabs HOA members can see.
2. HOA Super Users can see which tabs HOA Board members/Admins can see.
3. HOA Board/Admin can determine what should be the payment option for amenities, violation and late-fees.
4. HOA admin can set up late charge fees to be applied after someone is late by N number of days in paying their dues. So, when due payment happens, it should check if the

person is late in payment. If they are late, then it should add up that amount as a new line item in their checkout (late payment fee)

Section 15: News

1. HOA member should be able to see the news in the dashboard section
2. HOA Board/Admin should be able to add new news/updates from the news update section.

Section 16: Dashboard

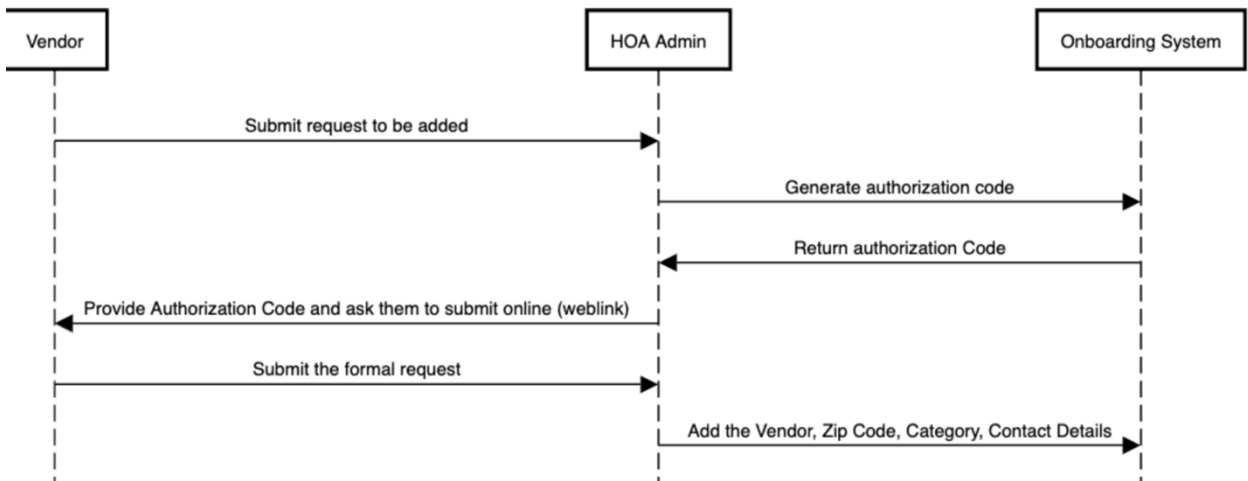
1. All HOA Members post-login, will land on the dashboard page.
2. User can be HOA Board or HOA Member. Such users should be able to view the tabs that are available as HOA Board such as community details, News, etc. (other option is to toggle the experience depending on the role that the user wants to take)
3. HOA Member should be able to see the summary view of upcoming action items such as upcoming payment info, service requests, News, Violations.

Section 17: vendor Management

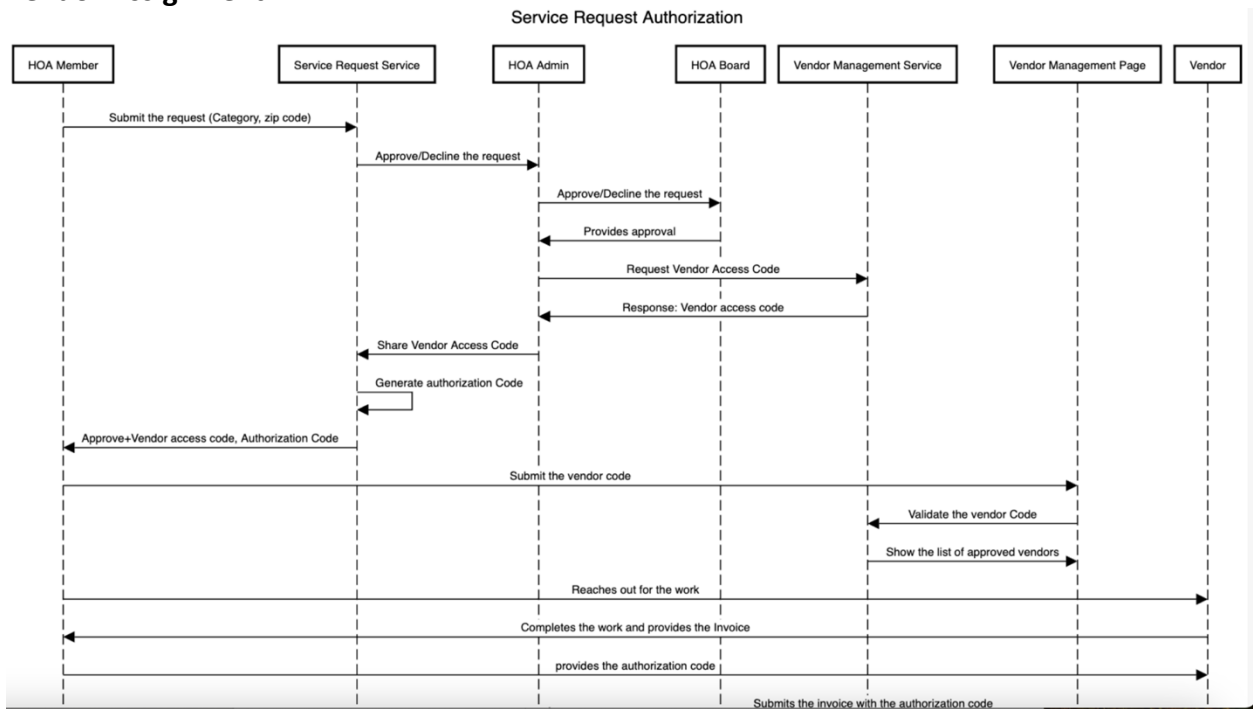
1. HOA Admin/Board should be able to onboard the vendors for their community. Onboarding Form link will be provided to the vendor or can be filled on behalf of the vendor.
2. HOA Admin/Board must maintain the list of vendors along with their license and insurance details (attachments)
3. Board members must have an option to generate ONE TIME USE ONLY & TIME CONSTRAINT vendor access code and contract code and attach those to the service request once it's approved.
4. HOA Board/Admin can assign vendors to a service request.
5. HOA Members should provide the vendor access code to see the list of authorized vendors.
6. HOA Members should provide the contract code to the vendor after the work is complete.
7. Members should be able to provide feedback about the vendor.

Vendor Onboarding Flow:

Vendor Onboarding



Vendor Assignment:



High level Ask:

1. External Developers will share the code as they complete each module.
2. Validate database design before starting the work.
3. Code must be modularized.
4. It should be able to support multiple different clients and thus HOA name, logo, address etc., should be customizable.
5. HOA Website will have an option for existing clients to login.

6. Our developers will determine the code quality and may suggest improvements.
7. All client data will be stored in AWS servers.
8. Website will support payment gateway PayPal. Should be able to accept Credit card, Debit card and Bank Account transfer.
9. Weekly sync is mandatory to determine the progress or address any blockers.
10. The functions should be written as per our design/workflows. Any deviation from that should be discussed with clients.
11. Data security is key. HOA Board members cannot see HOAs information. They can only see and edit their HOA related details. Bank details should be tokenized and stored in the vault. Payment MUST be PCI compliant.
12. Product will be primarily launch in US marketplace.
13. Front end – Reactjs.
14. Backend – Java springboard
15. Database – DynamoDB
16. Post launch support will be required.
17. Mock Link: <https://www.figma.com/file/jW28Pv7lXnVj2xeikgzLrK/HOA-%2F-Homeowner-V2?type=design&node-id=0%3A1&mode=design&t=tyWCGQja3KN19jxB-1>
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